Compass Calibration

The Seaglider computes heading from a 3-axis magnetometer plus a 3-axis accelerometer: the accelerometer gives pitch and roll, which are used to rotate the measured magnetic field into earth coordinates, and the horizontal field then points at magnetic north. Everything the glider does with that heading — steering, roll trim, depth-averaged currents, the smarter `$NAV_MODE`s — is only as good as the compass calibration, and IOP's guidance is blunt: **some form of compass calibration should be applied for every mission.**

> **Source**
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> Paraphrased from the APL-UW IOP *SGX Documentation* (§4.4.3), IOP webinar 3 (*Basestation3 CLI and customization + glider compass calibration*), and the UW *New compass calibration procedures for Seagliders* note (Bennett, 2012 — the "whirly" and in-flight procedures). Defer to APL-UW IOP for current tooling.

Hard iron, soft iron, and what the calibration does

The field the magnetometer measures is the sum of the earth's field and the glider's own magnetic signature:

- **Hard iron** — materials with their own permanent field (the steel in battery-cell casings is the classic one). It rides in the compass's own coordinate frame, so it appears as a constant **offset** on each axis. Usually — but definitely not always — the larger error source.
- **Soft iron** — materials that become magnetized in the presence of another field. Its effect changes with orientation, **stretching and rotating** the measurement.

Picture a glider tumbling through every orientation in a clean environment: the tip of the measured field vector sweeps out a **sphere centered on the origin** (the earth's field is constant in magnitude). Hard iron shifts that sphere off the origin; soft iron squashes it into an ellipsoid. Calibration is simply finding the transformation that turns the measured ellipsoid back into a sphere on the origin: an offset vector `(p, q, r)` for hard iron and a 3×3 matrix for soft iron.

Those numbers reach the glider through the `tcm2mat.cal` control file: